

SYZRAN, Russian Federation

WMO: 279830

Lat: 53.183N

Lon: 48.400E

Elev: 339

StdP: 14.52

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-9.8	-20.8	1.8	-15.1	-15.5	2.4	-9.1	18.7	20.6	16.2	15.5	3.8	260	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	89.4	67.3	85.7	66.6	82.3	65.5	70.0	82.9	68.6	81.4	67.2	79.4	6.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	96.1	75.4	64.2	91.0	74.1	62.7	86.4	72.5	34.2	82.8	32.9	81.5	31.8	79.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.3	13.6	DB	-21.9	94.0	6.2	3.6	-26.3	96.6	-30.0	98.7	-33.4	100.7	-37.9	103.2	(4)
(5)			WB	-21.9	72.8	5.8	1.0	-26.1	73.6	-29.5	74.2	-32.8	74.8	-37.0	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	13.7	14.3	26.0	44.6	57.6	66.0	70.1	66.7	56.4	43.0	29.9	18.7		
	DBStd	22.29	12.18	12.35	9.14	9.54	7.95	7.38	6.50	7.16	7.53	7.59	9.61	11.91		
	HDD50	4945	1124	999	744	218	25	1	0	0	21	239	603	971		
	HDD65	8692	1589	1419	1209	614	248	75	25	66	277	681	1053	1436		
	CDD50	2176	0	0	0	55	262	482	623	518	213	23	0	0		
	CDD65	448	0	0	0	1	20	107	183	119	18	0	0	0		
	CDH74	3977	0	0	0	19	226	945	1616	1030	141	0	0	0		
	CDH80	1318	0	0	0	1	30	312	565	383	27	0	0	0		
(14) Wind	WSAvg	6.5	6.2	6.6	6.7	7.1	7.2	6.6	6.1	6.1	6.1	6.6	6.8	6.5		
(15) Precipitation	PrecAvg	17.9	1.2	0.9	0.7	1.3	1.3	2.2	2.4	2.0	1.7	1.5	1.5	1.3		
	PrecMax	25.2	2.4	2.3	2.0	3.7	3.9	4.2	6.1	7.8	5.0	3.2	4.0	3.0		
	PrecMin	11.3	0.2	0.0	0.0	0.0	0.0	0.4	0.1	0.0	0.0	0.0	0.4	0.3		
	PrecStd	3.4	0.6	0.6	0.6	0.9	0.9	0.9	1.6	1.5	1.3	0.9	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	37.2	50.9	76.9	83.2	93.7	95.6	95.2	83.5	66.9	52.0	39.5		
		MCWB	35.0	34.2	43.2	55.8	61.0	67.0	67.8	66.9	61.7	53.7	49.2	37.8		
	2%	DB	34.8	35.4	43.5	70.7	79.7	87.4	89.3	88.5	78.0	61.9	48.0	36.6		
		MCWB	33.0	33.3	37.9	53.6	60.3	66.3	67.9	66.7	59.8	51.8	45.1	34.5		
	5%	DB	33.0	33.6	39.6	65.2	76.0	83.1	85.6	84.0	73.2	57.7	44.4	34.9		
		MCWB	31.5	32.0	35.4	51.2	59.4	65.8	67.4	65.9	59.0	50.5	42.2	33.1		
	10%	DB	31.0	31.6	36.8	60.2	72.0	79.2	82.4	79.4	68.4	54.3	41.5	33.3		
		MCWB	29.5	30.1	33.6	48.8	57.1	64.3	66.7	64.6	57.2	48.5	39.6	31.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	34.9	43.8	58.2	64.6	70.7	72.2	70.7	64.8	56.1	49.9	38.4		
		MCDB	36.1	36.1	50.9	70.8	76.5	84.1	84.8	84.4	76.6	61.6	51.7	39.3		
	2%	WB	33.1	33.5	38.5	55.1	62.7	68.5	70.7	68.7	62.2	53.5	45.6	34.9		
		MCDB	34.4	35.0	42.3	68.2	75.6	82.3	83.2	83.1	72.9	59.7	47.6	36.0		
	5%	WB	31.6	32.1	35.9	52.0	60.7	67.1	69.4	67.3	60.2	51.1	42.3	33.3		
		MCDB	32.9	33.6	38.9	64.1	72.6	80.0	81.6	81.1	69.6	56.9	44.0	34.5		
	10%	WB	29.5	30.0	34.0	49.3	58.7	65.5	68.0	65.7	58.2	49.1	39.7	31.9		
		MCDB	30.9	31.6	36.3	59.3	70.2	77.0	79.5	77.4	67.2	54.0	41.4	33.4		
(35) Mean Daily Temperature Range	MDBR		10.6	12.3	13.1	17.2	19.6	18.8	18.6	18.8	17.3	12.6	8.4	9.3		
	5% DB	MCDBR	8.0	9.4	14.6	25.3	26.2	25.1	24.1	26.4	24.6	20.0	10.8	7.4		
		MCWBR	7.7	8.2	10.2	12.5	11.2	9.6	8.6	9.6	10.4	11.3	8.9	7.1		
	5% WB	MCDBR	7.9	8.7	12.9	23.7	22.7	20.9	20.9	22.4	21.4	17.2	9.6	7.1		
		MCWBR	7.7	8.1	9.3	12.6	10.8	9.0	8.7	9.5	10.4	10.7	8.7	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.327	0.367	0.431	0.404	0.393	0.407	0.429	0.381	0.353	0.326	0.297		
	taud		2.082	2.039	2.033	2.088	2.286	2.363	2.321	2.244	2.350	2.340	2.238	2.101		
	Ebn at Noon		203	236	252	252	268	272	266	251	250	230	195	178		
	Edh at Noon		23	33	42	45	39	37	38	38	31	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		280	613	1069	1380	1782	1927	1905	1555	1024	521	222	153		
	RadStd		49	71	125	128	120	180	147	131	139	85	34	46		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+2.93	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+2.02	N/A	N/A	N/A	N/A	+167	N/A

Nomenclature: See separate page